

Blowup profiles for the Okamoto, Sakajo and Wunsch family: exact stagnation point relations and the regularity exponent

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Abstract

We study finite time blowup in the one parameter family of one dimensional models for the three dimensional Euler vorticity equation introduced by Okamoto, Sakajo and Wunsch, which interpolates between the Constantin, Lax and Majda model at $a = 0$ and the De Gregorio model at $a = 1$. On the circle, Lushnikov, Silantyev and Siegel identified two regimes separated by a critical advection $a_c = 0.6890665\dots$: below it the blowup narrows, while above it the solution converges to a fixed profile f with $\omega = f(x)/(T - t)$, whose Fourier spectrum decays algebraically because a high order derivative of f jumps at the point antipodal to the singularity. They obtained the exponent of that decay by fitting the spectrum. This paper supplies a local theory for the frozen profile which predicts the exponent instead of fitting it.

Rearranging the profile equation shows that f can be singular only where its own velocity vanishes. There are exactly two such stagnation points, they are exactly π apart, which we show is forced by a parity argument, and at each the local exponent is $\mu = (c - 1)/(ac)$ where c is the Hilbert transform of f at that point. Since $|y|^\mu$ has Fourier coefficients decaying like $k^{-(\mu+1)}$, one local number therefore predicts the entire spectral decay rate $\beta = \mu + 1$. At the stagnation point where f has a simple zero this closes in elementary form as $c = 1/(1 - a)$, verified to machine precision; the general self similar version, $h = (1 + c_l)/(1 - a)$, also appears in recent work of Xu, and an equivalent balance gives the complex singularity exponent of Lushnikov, Silantyev and Siegel. The other stagnation point is the one that carries the singularity, and its c is genuinely global: no local balance determines it. Using the exact value at the first point as a control variate removes the discretisation error at the second, giving five digits or better. We conclude that β varies continuously with a and is not equal to 3, with $\beta = 3.0024227$ at $a = 0.8$, and that at $a = 0.71$ the predicted $\beta = 9.2955$ sits 0.33 percent below the published fitted value 9.3259.

We add three further results. The linearisation about the profile, taken in renormalised time, has exactly two eigenvalues outside the open left half plane, both forced by symmetry, so the profile is linearly stable modulo those symmetries, which sharpens the nonlinear stability previously inferred from convergence in simulations. The profile satisfies a fixed point equation $\mathcal{R}(f) = f$ admitting no free scaling, so it determines its own amplitude, and that amplitude agrees with the simulation to 10^{-5} . Finally we give an exact first integral of the profile equation and a global constraint, $\text{PV} \oint dy/U = 0$, verified to 10^{-12} , and we record that at $a = 1/2$, where the model is exactly solvable, the datum $\omega_0 = \sin x$ blows up at $T = \pi$ exactly.

1 Introduction

The question of whether solutions of the three dimensional incompressible Euler equations can develop singularities from smooth data remains open, and the same question for Navier and Stokes is one of the Clay Millennium Prize problems. A recurring strategy is to study one dimensional models which retain the competition between vortex stretching and transport while discarding the geometry.

Constantin, Lax and Majda [1] introduced the model

$$\omega_t = \omega \mathcal{H}\omega, \quad (1)$$

where $\mathcal{H}$ is the Hilbert transform, and solved it in closed form. De Gregorio [2] restored the transport term, and Okamoto, Sakajo and Wunsch [3] embedded both in the one parameter family

$$\omega_t + a u \omega_x = \omega u_x, \quad u_x = \mathcal{H}\omega, \quad (2)$$

posed here on the circle with u of zero mean. The parameter a tunes the strength of transport against stretching. At $a = 0$ transport is absent and (2) reduces to (1); at $a = 1$ the two terms are in genuine competition, and this case is understood to be critical. Jia, Stewart and Šverák [4] studied the ground state of the De Gregorio model, Lei, Liu and Ren [5] proved global regularity on the circle for data of one sign, and Chen, Hou and Huang [6] proved finite time blowup on the line for data of low Hölder regularity. The broader programme of computer assisted singularity detection in fluid equations is exemplified by Luo and Hou [7], Elgindi [8] and Chen and Hou [9].

Two features of (2) make it a good testbed. Both endpoints of the family are known exactly, which makes every numerical statement checkable, and the solution structure is simple enough that the blowup profile can be computed directly rather than inferred from a diverging simulation.

Relation to existing work. Self similar blowup for this family is under active study and the present results must be read against it.

The pole dynamics line. The closest work to this paper is a body of results obtained by complex singularity and pole dynamics methods, and it already contains much of the setting used here. Lushnikov, Silantyev and Siegel [17] sweep a on both the real line and the circle with high precision spectral simulations and a nonlinear eigenvalue solve. They locate the critical advection $a_c = 0.6890665337007457\dots$ at which the width exponent c_l vanishes; they identify, for $a_c < a \lesssim 0.95$ on the circle, the frozen blowup $\omega = f(x)/(T - t)$ which is the object of this paper, and compute its profile for $a_c < a \leq 0.85$; they observe that f has a jump in a high order derivative at $x = \pm\pi$, antipodal to the singularity, which at $a = 0.8$ is a jump in f'' ; and they measure the resulting algebraic spectral decay, reporting its exponent as p_b , which is our β . Their Theorem 1 gives the exponent $\gamma = 1/(1 - a)$ of the leading complex singularity, from the balance $a\gamma = \gamma - 1$, which is the same algebra as Proposition 8 applied to a different object. We claim none of this, and the regime diagram of Section 4 should be read as reproducing theirs rather than as establishing it.

What this paper adds to that is the local theory. Their p_b comes from a least squares fit to the spectrum over a window; ours follows from (15) given a single value of $\mathcal{H}f$, and Section 10 shows that the two differ in the third digit.

Silantyev, Lushnikov, Siegel and Ambrose [18] construct exact pole dynamics solutions on the circle for $a = 0$ and $a = 1/2$; Section 9 is an application of theirs and claims no more. Xu [19] studies the spectrum of the linearisation about the collapse profile at $a = 0$ on the real line, obtaining the essential spectrum, a point spectrum $\{0, 1\}$ consisting of symmetry modes, and a spectral gap. His Section 2 records the general self similar profile equation (12), and in the course of his Section 4 he derives $\mathcal{H}\Omega(0) = (1 + c_l)/(1 - a)$, which is (14) below; Proposition 7 is therefore his, obtained independently here, and we retain it only because Proposition 6 needs it. Our Section 7 asks his question for the frozen profile on the circle, where the second symmetry mode is translation rather than dilation, and answers it numerically where he answers it with proof.

Okamoto, Sakajo and Wunsch [3] study global existence against blowup using a criterion in the spirit of Beale, Kato and Majda together with energy estimates, and do not consider self similar profiles. They do fit the spectrum in the combined form $|\hat{\omega}_n| \sim Cn^{-p}e^{-\delta n}$ and track both parameters in time, observing that $\delta(t)$ decays exponentially, which they read as evidence of smoothness for all time, while $p(t)$ decreases monotonically and, in their words, they are unable to see its asymptotic value from the numerical data. The exponent β of (15) is the analogue of that p for a frozen blowup profile, and (15) determines it in closed form given one local quantity.

Huang, Tong and Wei [11] construct infinitely many compactly supported self similar blowup solutions of the De Gregorio model on the real line, organised as eigenfunctions of a compact operator and as solutions of nonlinear eigenvalue problems, with profiles in $H_0^1 \cap H^2$ and smooth in the interior of the support. Huang, Tong and Wang [16] study self similar blowup of the generalised model from data with derivative degeneracy, reporting one scale blowup with regular profiles for $a > 0$ against a two scale scenario with singular outer profile for $a \leq 0$, and constructing an explicit family of singular profiles in the latter regime. Both use the standard self similar ansatz, whose profile equation $(c_l x + au)\omega_x = (c_\omega + u_x)\omega$ reduces to (7) at $c_l = 0$, $c_\omega = -1$. We claim no novelty for that formulation.

Huang, Qin, Wang and Wei [12] prove that the family admits exact self similar blowup with interiorly smooth profiles for all $a \leq 1$, obtained as a fixed point of an a dependent nonlinear map, with characterisations of regularity, monotonicity and far field decay. Their profiles fail to be smooth only at the boundary of a compact support, whereas the profile here is $C^{1,1}$ at an *interior* point. The two are consistent, and the reason is the domain: on the circle U has zero mean by construction, so it must change sign and interior stagnation points are forced, while on the line they are not. Their analogues of c_1 and c_2 are defined as integrals and bounded rather than evaluated, for instance $2\eta/\pi \leq c(f) \leq 4/\pi$, and no closed form appears.

Huang, Qin and Wang [13] construct blowups with two collapsing spatial scales on the line, one measuring the distance to the blowup point and one the shrinking bulk. Those profiles narrow, $c_l > 0$, so they belong to the narrowing regime of Section 4 rather than the frozen one, and the blowup there occurs where ω_0 vanishes with $\mathcal{H}\omega_0 > 0$ rather than at a zero of the velocity.

Guo and Jiu [15] study stability on the torus at $a = 1$ near the excited state $-\sin 2\theta$, proving linear and nonlinear instability for a broad class of data. That is the steady state analogue, at the critical parameter, of the observation in Section 6 that the higher members of Proposition 4 are not generic attractors. It concerns equilibria rather than blowup profiles and takes no renormalised time limit.

Zheng [14] constructs exactly self similar C^α blowup for the generalised De Gregorio equa-

tion at small $|a\alpha|$, again on the line and again narrowing, with rescaling factor $(1-t)^{1+\lambda(a)}$.

A pattern runs through the constructive part of this literature. With the exception of [17] and [18], every construction we are aware of works on the real line with a narrowing profile, $c_l \neq 0$. The circle behaves differently for a reason that is not a matter of taste: there U has zero mean by construction, so it must change sign, interior stagnation points are forced, and by Proposition 6 the profile cannot be smooth at them. That is the setting in which a frozen profile, $c_l = 0$, becomes the attractor, and it is the setting [17] already works in.

What remains to be done there is the local analysis at those stagnation points. Proposition 6 at the *singular* stagnation point, where f has no simple zero and $\mu \neq 1$, is the part that none of these references contains, and it is what turns the fitted exponent of [17] into a predicted one. The parity statement fixing the two stagnation points π apart, the control variate of Section 10, the first integral and global constraint of Section 11, and the linear stability computation of Section 7 are likewise not in them. Two numerical coincidences with [16] invited closer comparison, and Proposition 7 is what makes that comparison possible, since it puts both settings in one formula.

The transition near $a \approx 0.75$. [16] identifies $a^{c,2} \approx 0.751$ as the value at which its self similar profile *loses* stability as a increases. The frozen profile studied here *gains* selection as a increases through roughly the same value. These are not the same object: their profiles carry $c_l \neq 0$ and therefore narrow, which in the language of Section 4 places them in the narrowing regime, whereas a frozen profile requires $c_l = 0$ exactly. It is tempting to read the two as complementary halves of one exchange of stability, in which the narrowing attractor gives way to the frozen one.

Table 2 tests that reading. The mechanism survives and the location does not.

The mechanism: c_l falls smoothly to zero rather than jumping, so the narrowing branch merges with the frozen one rather than being replaced by it, which is the right shape for an exchange.

The location: an earlier version of this paper read Table 2 as giving $c_l = 0.004536 \pm 0.000543$ at $a = 0.751$, eight standard errors from zero, and concluded $a_c \gtrsim 0.795$. That is withdrawn, and it was wrong. The critical advection is known to far better precision than anything measured here: [17] obtains $a_c = 0.6890665337007457 \dots$ from a nonlinear eigenvalue problem, and [19] recomputes the branch independently by Newton continuation, finding its zero crossing at 0.6888, in agreement to 0.04 percent.

The failure is one of resolution, and it is visible in Table 2 itself once the published branch is put beside it: 1.058 against 1.0000 at $a = 0$, 0.748 against 0.7474 at $a = 0.2$, 0.512 against 0.4809 at $a = 0.4$, 0.325 against the exact $1/3$ at $a = 0.5$, and 0.179 against 0.1691 at $a = 0.6$. A half width measurement carrying several percent of bias cannot resolve a quantity of size 0.0045, and fitting $c_l = A(a_c - a)^p$ through it, with the best a_c running from 0.795 at $p = 2$ to 0.844 at $p = 4.5$, was an extrapolation through convex data of exactly the kind warned against in Section 10.

An independent check from this paper agrees with the published value rather than with the withdrawn one. [17] reports the spectral exponent diverging as $a \rightarrow a_c^+$, and β as determined in Section 10 does the same, taking the values 16.362, 9.295, 6.795 and 4.213 at $a = 0.70, 0.71, 0.72, 0.75$. Extrapolating $1/\beta$ linearly to zero from the two closest pairs gives 0.6868 and 0.6828, with the pair nearer the transition giving the larger value, consistent with $a_c = 0.68907$ and not with 0.795. We quote this loosely, since $1/\beta$ is convex and this is the

same trap once more, but its direction is not in doubt.

The coincidence with [16] therefore stands unresolved rather than refuted, and our earlier reason for refuting it was itself an artefact. Their $a^{c,2} \approx 0.751$ is a stability boundary for a profile on the line, while $a_c = 0.68907$ is where the width exponent vanishes on the circle. The two numbers are close and describe different events, and nothing established here decides whether they are related.

The constant $1/(1-a)$. Theorem 2.6 of [16] gives, for $a < 0$, an exact solution $\bar{\Omega}_a(X) = -\sin(\pi\mu) \mathbf{1}_{X>1} |1-X|^{-\mu}$ with $\mu = 1/(1-a)$ and scaling factors $c_{l,a} = 1-a$, $c_{\omega,a} = -1$. The resemblance to Proposition 8 is superficial. Their μ is an exponent and equals $1/c_{l,a}$ in their normalisation; ours is a value of $\mathcal{H}f$ at a point where it is finite. Equation (13) does not even apply at their singular point, since $\mathcal{H}\bar{\Omega} = 2/(1-X^2)$ diverges there and the advection velocity does not have a simple zero. What the two share is the algebraic origin of the factor: $(1-a)$ is what appears whenever the stretching coefficient 1 is balanced against the advection coefficient a , entering our case through $h(1-a) = 1+c_l$ and theirs through the scaling normalisation. Neither statement implies the other.

The same factor appears twice more in the pole dynamics literature, and once by the same computation. Theorem 1 of [17] gives $\gamma = 1/(1-a)$ for the power of the leading complex singularity of ω off the real axis, from the balance $a\gamma/(\gamma-1) = 1$, that is $a\gamma = \gamma-1$, which is identical in form to the $ac = c-1$ of Proposition 8. The objects are different, theirs a power at a complex singularity and ours a value of $\mathcal{H}f$ at a real stagnation point, but the balance of stretching against advection which produces the factor is the same, and we make no priority claim for it. More directly, [19] states the general form $\mathcal{H}\Omega(0) = (1+c_l)/(1-a)$ outright, which is Proposition 7, of which Proposition 8 is the $c_l = 0$ case.

Our contributions are as follows.

1. We recover the two regime picture of [17] on the circle with an independent and translation invariant diagnostic, and record where our resolution falls short of theirs (Section 4).
2. We formulate the frozen profile equation as a fixed point of the evolution nonlinearity, observe that it admits no free scaling and therefore fixes the profile amplitude outright, and verify that prediction against simulation to 10^{-5} (Section 5).
3. We compute the spectrum of the linearisation about the frozen profile in renormalised time and identify the two symmetry forced eigenvalues exactly, one of which is provable in closed form, which sharpens the nonlinear stability inferred in [17] from observed convergence (Section 7).
4. We prove that singularities of the profile occur only at stagnation points of its velocity, that there are exactly two and that they are exactly π apart, and we derive the local exponent $\mu = (c-1)/(ac)$ there. The closed form $c = 1/(1-a)$ at the stagnation point where f is smooth is the $c_l = 0$ case of a relation also given in [19]; the exponent at the *singular* stagnation point, where $\mu \neq 1$, is the one the rest of the paper rests on (Section 8).
5. We determine the remaining, global constant to five digits using the exact relation as a control variate, concluding that the spectral decay exponent varies continuously with

a , that it is not equal to 3, and that at $a = 0.71$ it sits 0.33 percent below the value obtained in [17] by fitting the spectrum (Section 10).

6. We test the c_l dependence of Proposition 7 against the exact $a = 1/2$ solution of [18], and record that the datum $\omega_0 = \sin x$ blows up there at $T = \pi$ exactly (Section 9).
7. We give a first integral of the profile equation and a global constraint which we verify numerically (Section 11).

Throughout, the initial datum is $\omega_0 = \sin x$. This is not a generic choice, and Section 13 records what that costs.

2 The model and its exactly known endpoints

Write ω for the vorticity on the circle $x \in [0, 2\pi)$ and let $\mathcal{H}$ denote the Hilbert transform, acting on Fourier modes as $\widehat{\mathcal{H}\omega}_k = -i \operatorname{sgn}(k) \hat{\omega}_k$. The velocity is recovered from $u_x = \mathcal{H}\omega$ with u of zero mean, so $\hat{u}_k = -\hat{\omega}_k/|k|$ for $k \neq 0$.

The case $a = 0$. Setting $z = \mathcal{H}\omega + i\omega$ converts (1) into the Riccati equation $z_t = z^2/2$, whence

$$\omega(x, t) = \frac{4\omega_0(x)}{(2 - t\mathcal{H}\omega_0(x))^2 + t^2\omega_0(x)^2}. \quad (3)$$

For $\omega_0 = \sin x$ the denominator collapses to $4 + 4t \cos x + t^2$ and

$$\|\omega(t)\|_\infty = \frac{4}{4 - t^2}, \quad (4)$$

so the blowup time is $T = 2$ and the rate exponent is exactly one.

The case $a = 1$. For $\omega_0 = \sin x$ one has $\mathcal{H}\omega_0 = -\cos x$ and $u = -\sin x$, so $u\omega_x = \omega u_x$ pointwise and $\sin x$ is a steady state. Hence $T = \infty$.

The family therefore runs between a case whose blowup time is exactly 2 and a case whose blowup time is exactly infinite, and any numerical scheme can be calibrated against both.

3 Numerical method

Space is treated pseudospectrally: products are formed on an equispaced grid of N points, dealiased by the two thirds rule, with the Nyquist mode set to zero. Time stepping is classical fourth order Runge and Kutta, with the step governed by two constraints. Accuracy near blowup requires $\Delta t = \text{cfl}/\max(\|u_x\|_\infty, \|\omega\|_\infty)$, since the stretching term drives the growth. Stability requires $\Delta t < c_{\text{adv}}/(k_{\text{cut}} |a| \|u\|_\infty)$, because the transport term has eigenvalues on the imaginary axis out to $i a u k_{\text{cut}}$ and the method is stable there only to 2.828.

Two diagnostics require care. First, the supremum norm is computed from the spectral interpolant rather than the grid samples: the grid maximum errs by $O(\Delta x^2 f'')$, and since the peak narrows like $1/\|\omega\|_\infty$ near blowup, that error grows precisely when it matters. We locate the peak by Newton iteration on $f'(x) = 0$ against the interpolant, at $O(N)$ cost for a

single point. Second, spectral decay rates are measured over a band bounded away from both ends: the low wavenumber end is contaminated by the analytic factor multiplying the singular behaviour, and the high wavenumber end by the dealiasing mask. Section 10 quantifies both.

The implementation passes nineteen checks against closed forms and conserved quantities, including (3) and (4) to 2×10^{-11} , conservation of $\int \omega$ to 10^{-16} for $a \in \{0, \frac{1}{2}, 1, -\frac{1}{2}\}$, spatial convergence following the predicted analyticity strip across four orders of magnitude, and fourth order temporal convergence with measured ratios 16.65, 16.28, 16.16, 16.07.

4 Blowup times and two distinct regimes

From $\omega_0 = \sin x$ we observe finite time blowup for every $a < 1$, with rate exponent one throughout. Table 1 collects the blowup times, estimated by fitting $\omega/(d\omega/dt) = (T - t)/p$, a straight line in t with slope $-1/p$ and root T .

a	0	0.2	0.4	0.5	0.6	0.7	0.8	0.9	1
T	2 (exact)	2.3101	2.7858	π (exact)	3.6500	4.4641	6.0719	10.757	∞ (exact)
p	1 (exact)	0.997	1.000	1.001	1.001	0.999	1.000	1.049	n/a

Table 1: Blowup time and rate exponent from $\omega_0 = \sin x$. No transition to global existence appears below $a = 1$; instead T diverges as $a \rightarrow 1$.

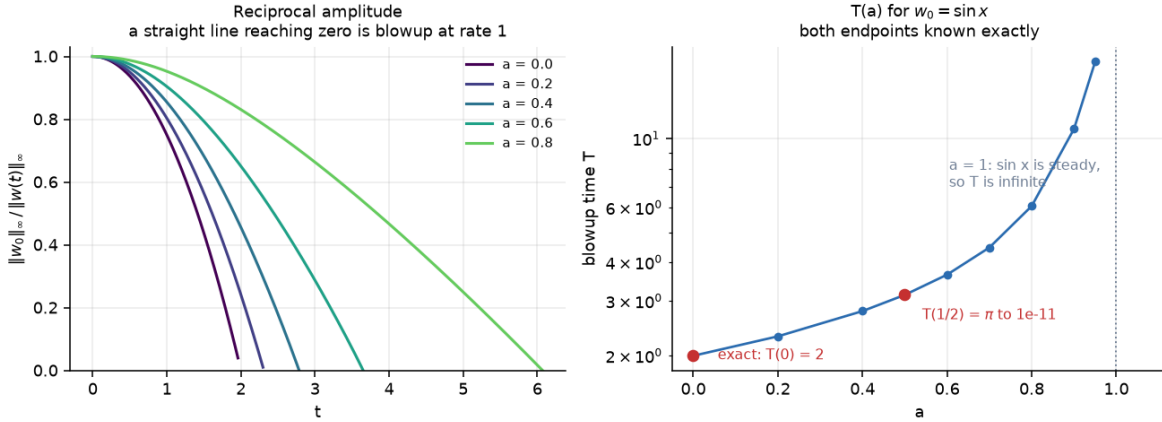


Figure 1: Left: reciprocal amplitude $\|\omega_0\|_\infty / \|\omega(t)\|_\infty$ for several values of a . A straight line reaching zero is blowup with rate exponent one. Right: the blowup time $T(a)$ from $\omega_0 = \sin x$, on a logarithmic scale. Both endpoints are known exactly, $T(0) = 2$ and $T(1) = \infty$, and the curve runs between them without any transition to global existence at intermediate a . The agreement of $T(1/2)$ with π is marked but unexplained.

Remark 1. The value at $a = 1/2$ agrees with π to 1.2×10^{-11} and continues to do so under refinement across an eightfold range in N . We have no explanation for this and record it as an observation.

The blowups are not all alike. Comparing normalised spectra $|\hat{\omega}_k| / \max_k |\hat{\omega}_k|$ at successive decades of amplitude growth gives a diagnostic which is invariant under translation and assumes nothing about the profile shape. At $a = 0.8$ the curves converge: successive deviations from the first snapshot are 3.30×10^{-3} , 3.68×10^{-3} , 3.76×10^{-3} , 3.78×10^{-3} across four decades of growth, the peak location locks to $x = 2.16699$, and the spectral tail remains at 2.7×10^{-17} throughout, so a grid of 4096 points carries an amplitude of 10^6 without strain. At $a = 0.4$ the same comparison moves by 0.72 over a single decade and resolution is exhausted by $\|\omega\|_\infty = 5.8 \times 10^3$.

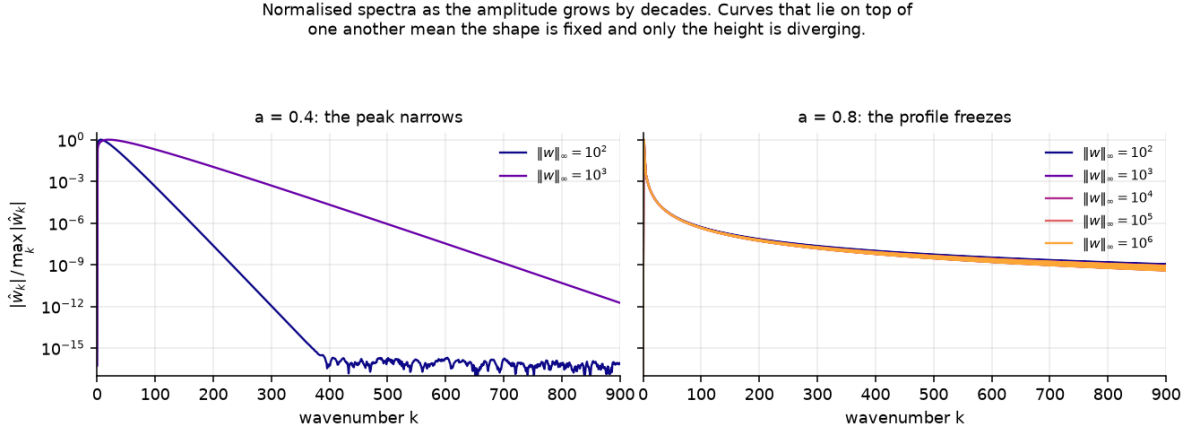


Figure 2: Normalised spectra $|\hat{\omega}_k| / \max_k |\hat{\omega}_k|$ at successive decades of amplitude growth. Taking the modulus removes the translation, so no alignment of the peaks is required and the comparison assumes nothing about the profile shape. Left, $a = 0.4$: the curves spread steadily as the peak narrows, and resolution is exhausted by $\|\omega\|_\infty = 5.8 \times 10^3$. Right, $a = 0.8$: five curves spanning four decades of growth lie on top of one another, so the shape is fixed and only the height diverges.

We refer to the first behaviour as a *frozen* profile and the second as *narrowsing*, illustrated in Figure 2. Only the frozen case is analysed below.

The dichotomy is due to [17], who established it on the circle, located the transition at $a_c = 0.6890665337007457\dots$, and computed the width exponent to five to eight digits by a nonlinear eigenvalue solve. What this section reports is an independent measurement of the same quantity by a different and much cruder route, and it should be read as a check on our diagnostic rather than as a determination of theirs.

The distinction is quantitative. In the general self similar ansatz of Proposition 7, the peak width scales like $(T - t)^{c_l}$ while the amplitude grows like $(T - t)^{-1}$, so width $\sim$ amplitude $^{-c_l}$ and c_l is minus the slope of one against the other on logarithmic axes. A frozen profile is exactly $c_l = 0$. Measuring the width as the half width at half maximum around the peak, which is robust whether the spectrum is exponential or algebraic, gives Table 2. The method is calibrated at $a = 0$, where the exact solution (3) has width proportional to $(2 - t)$ and amplitude $4/(4 - t^2)$, so $c_l = 1$; the measurement returns 1.048. Against the published branch, Table 2 is accurate to a few percent and not of one sign; the introduction gives that comparison term by term, together with the estimate of a_c it forces us to withdraw.

Remark 2 (One signed data). Lei, Liu and Ren [5] proved global regularity on the circle for data of one sign at $a = 1$. The mechanism appears to reach well below that. From

a	0	0.2	0.4	0.5	0.6
c_l	1.058	0.748	0.512	0.325	0.179
a	0.70	0.74	0.75	0.77	0.80
c_l	0.032321	0.007273	0.004536	0.001294	0.000398
std. err.	1.5×10^{-3}	7.5×10^{-4}	5.4×10^{-4}	3.0×10^{-4}	1.2×10^{-4}

Table 2: The spatial scaling exponent measured from the dynamics. The exact value at $a = 0$ is 1. The upper block is resolution limited and quoted to two figures; the lower block is fitted over three decades of amplitude, to 10^6 . Values at $a \geq 0.78$ are omitted because they are not monotone in a at the 5×10^{-4} level, which is the true noise floor rather than the smaller figure the regression reports.

$\omega_0 = 1 - \cos x$, integrating to $t = 200$, the amplitude remains bounded at 2.72, 2.40 and 2.17 for $a = 0.7, 0.8$ and 0.9 , while for $a = 0.6$ the solution reaches 10^6 by $t = 133$ and for $a = 0.5$ it exhausts resolution at 9.6×10^3 . The threshold lies between 0.6 and 0.7. We record this as an observation; it says nothing about the frozen profile, which is reached from sign changing data.

5 The profile equation

Substituting the ansatz

$$\omega(x, t) = \frac{f(x - x_p)}{T - t} \quad (5)$$

into (2) gives $f/(T - t)^2$ on the left and $\mathcal{R}(f)/(T - t)^2$ on the right, where

$$\mathcal{R}(f) = f \mathcal{H}f - a U f', \quad U' = \mathcal{H}f, \quad \int U = 0 \quad (6)$$

is the same quadratic nonlinearity that drives the evolution. The profile equation is therefore the fixed point problem

$$\boxed{\mathcal{R}(f) = f}. \quad (7)$$

Proposition 3 (No free scaling). *If f solves (7) and $c \neq 1$, then cf does not.*

Proof. $\mathcal{R}$ is quadratic, so $\mathcal{R}(cf) = c^2 \mathcal{R}(f) = c^2 f$, which equals cf only when $c^2 f = cf$, that is $c = 1$. $\square$

Proposition 3 is what makes (7) useful rather than merely descriptive: unlike a linear eigenvalue problem it fixes the amplitude of f outright. Since (5) gives $\|\omega(t)\|_\infty = \|f\|_\infty / (T - t)$, we obtain the falsifiable prediction

$$\frac{d}{dt} \frac{1}{\|\omega\|_\infty} = - \frac{1}{\|f\|_\infty}. \quad (8)$$

The left side is measured by time marching the partial differential equation; the right side by Newton's method on an algebraic fixed point. The two share nothing but the value of a .

We solve (7) by Newton iteration, assembling the Jacobian one column at a time through the same masked transforms used to evaluate $\mathcal{R}$, and regularising the translation mode by a

rank one shift along f' . At $a = 0.8$ the profile taken directly from the simulation satisfies (7) to 1.3×10^{-5} before Newton is applied; the iteration then converges quadratically, with residuals 1.35×10^{-5} , 4.07×10^{-11} , 1.77×10^{-12} . Table 3 compares the two sides of (8).

N	simulation, via (8)	profile equation	agreement
2048	4.148459148	4.148481857	5.5×10^{-6}
4096	4.154779433	4.154737187	1.0×10^{-5}

Table 3: $\|f\|_\infty$ at $a = 0.8$, computed two independent ways.

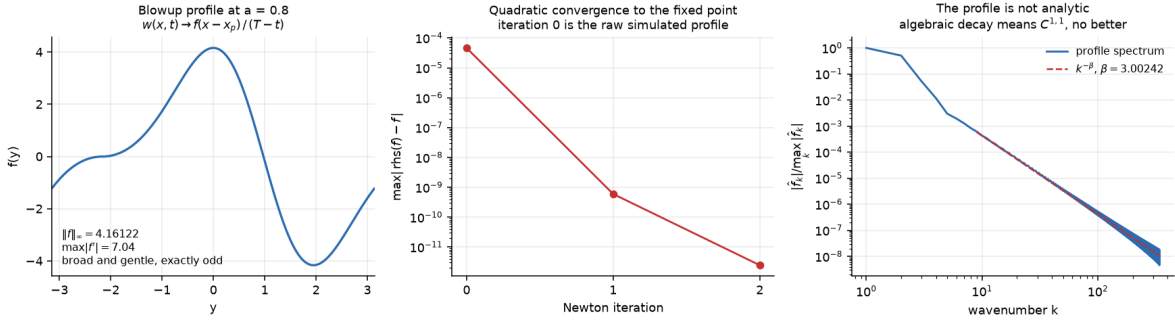


Figure 3: The blowup profile at $a = 0.8$. Left: $f(y)$, broad and gentle, with $\|f\|_\infty = 4.16$ against $\max|f'| = 7.04$, and exactly odd about each stagnation point. Centre: the Newton residual for (7), starting from the profile taken directly out of the simulation, which already satisfies the equation to 1.3×10^{-5} ; convergence is quadratic. Right: the profile spectrum on logarithmic axes against $k^{-\beta}$ with $\beta = 3.00242$ from (15), the exponent predicted in Section 10 rather than fitted to this curve. The decay is algebraic, not exponential, so the profile is not analytic.

The shared value drifts by 1.5×10^{-3} between the two resolutions, which is not Newton finding different branches: re-solving at $N = 4096$ from the interpolated $N = 2048$ solution returns the same answer, and mapping it back down reproduces the $N = 2048$ value to 2×10^{-12} . Each grid has one solution and the simulation finds that grid's solution. The slow convergence is explained in Section 8: the profile is only $C^{1,1}$.

Proposition 4 (Dilation). *If f solves (7) then so does $f(n \cdot)$ for every positive integer n , with the same $\|f\|_\infty$.*

Proof. The Hilbert transform commutes with integer dilation on the circle, so $\mathcal{H}[f(n \cdot)](x) = (\mathcal{H}f)(nx)$. Since the velocity is recovered by integrating $U' = \mathcal{H}f$, the velocity of $f(n \cdot)$ is $U(nx)/n$, while $\frac{d}{dx}f(nx) = nf'(nx)$. The two factors of n cancel in the advection term, so

$$\mathcal{R}(f(n \cdot))(x) = f(nx)(\mathcal{H}f)(nx) - a \frac{U(nx)}{n} nf'(nx) = [f \mathcal{H}f - a U f'](nx) = f(nx). \quad \square$$

So profiles come in families. The n th member carries $2n$ stagnation points spaced π/n , with exponents alternating between the two values of the base member, and $\|f\|_\infty$, c_1 , c_2 and β are all unchanged. Numerically, $f(n \cdot)$ solves (7) to 10^{-11} for $n \leq 4$, against 2×10^{-12} for the base profile itself.

Genericity. Every result here starts from $\omega_0 = \sin x$, which is the ground state of the $a = 1$ model and therefore not a generic datum. Proposition 5 and the stability computation of Section 7 predict that any datum in the basin converges to the same profile, and this is testable independently of them. At $a = 0.8$ and $N = 2048$:

datum	$\ f\ _\infty$	c_1	c_2	β
$\sin x$	4.1484819	4.996663	-1.6580199	3.003911
$\sin x + 0.3 \sin 2x$	4.1484819	4.996663	-1.6580199	3.003911
$\sin x + 0.5 \cos 2x - 0.2 \sin 3x$	4.1484819	4.996663	-1.6580199	3.003911
$\sin 3x + 0.4 \sin 4x$	4.1612203	5.006926	-1.668103	2.999354

Table 4: The same profile is reached from every datum tested. The first three agree to seven digits. The fourth lands on the $n = 2$ member of Proposition 4, which on a grid of 2048 points is represented as accurately as the base member on 1024 points, and indeed reproduces the $N = 1024$ values of Table 7 exactly.

Selection. Solutions of (7) exist over a wider range of a than the dynamics selects. Starting Newton from each simulation’s own profile and recording the residual *before* the iteration begins gives an order parameter: it reads 0.33 to 0.56 for a between 0.3 and 0.6, then 0.19, 5.0×10^{-2} , 2.6×10^{-3} , 1.3×10^{-5} and 5.5×10^{-5} at $a = 0.65, 0.70, 0.75, 0.80, 0.85$. The frozen regime is therefore roughly $0.75 \leq a \leq 0.85$ at the amplitudes reached, degrading smoothly rather than switching off.

6 The basin, and a third outcome

Since $\mathcal{R}$ is quadratic, replacing ω_0 by $\lambda\omega_0$ rescales time and changes nothing else, so the basin is scale invariant and only shape matters. Combined with Proposition 4, the question is which member a datum reaches.

Member one is the generic attractor. Sixteen random data, modes 1 to 8 with amplitudes falling like $1/k$ and random phases, carrying between 4 and 10 sign changes, all reach member one with $\|f\|_\infty = 4.148482$ in every case. The number of sign changes does not select the member. Higher members are reached only from data with the matching exact symmetry: $\sin(kx)$ gives member k for $k \leq 4$, and since $\sin(kx)$ is $2\pi/k$ periodic it remains in that symmetric subspace, while any generic perturbation returns the solution to member one.

A topological boundary. For $\omega_0 = \sin x + m$, which changes sign for $m < 1$ and is one signed at $m = 1$, blowup occurs for every $m < 1$ with $\|f\|_\infty = 4.148482$ throughout and T growing as 6.07, 10.07, 15.52, 20.28, 25.88, 43.81 at $m = 0, 0.5, 0.8, 0.9, 0.95, 0.99$, and ceases at $m = 1$ exactly, where the sign change degenerates into a touching zero. We fitted the divergence of T and report no law: neither $-\log(1 - m)$ nor $(1 - m)^{-p}$ has stable constants over this range.

c	0.46	0.475	0.48	0.50	0.52	0.56	0.58
outcome	member 1	member 1	member 1	decay	decay	member 2	member 2
T	61.4	90.4	108.2	-	-	54.9	44.0

Table 5: T diverges on approach from both sides. Inside the window the amplitude falls monotonically: at $c = 0.5$ it reaches 7.4×10^{-3} by $t = 400$ with $\|\omega\|_\infty t$ settling on 3.09, 3.03, 2.99, 2.97, so the solution decays like $1/t$.

A third outcome. Sign change is not sufficient. For $\omega_0 = \sin x + c \sin 2x$ the two members compete, and between them lies a window in which the solution does neither:

The window is open rather than a separatrix, since both $c = 0.50$ and $c = 0.52$ fail to blow up while 0.48 and 0.56 do, so the model admits global decaying solutions from an open set of sign changing data. The window is narrow, roughly $0.49 < c < 0.55$.

Decay like $1/t$ invites the guess that the interior is self similar. If $\omega \sim g/t$ then $R(g) = -g$, and since $\mathcal{R}$ is quadratic $\mathcal{R}(-g) = \mathcal{R}(g)$, so $h = -g$ would solve (7): the attractor would be a profile entered with the opposite sign, of amplitude 2.95, matching neither member. Writing $s = \log t$ and $W = t\omega$ gives $W_s = \mathcal{R}(W) + W$, whose linearisation is $-J$, so such a profile would have to be strongly unstable as a blowup profile.

It is not so. Two tests refute it. Newton started from $-t\omega$ at $t = 400$ returns member 1, with $\|f\|_\infty = 4.1484819$ and its constants and spectrum; the guess carried an eight percent residual and was never near a solution. And integrating to $t = 2000$ separates the amplitude from the shape: the decay exponent converges to 1 and $\|\omega\|_\infty t$ to 2.950, while the normalised spectrum drifts by 4.7×10^{-3} , 1.9×10^{-2} , 4.6×10^{-2} at $t = 400, 1400, 2000$ and the residual of $-t\omega$ grows from 6×10^{-2} to 4.9 with its sup norm unchanged. Fine structure is accumulating at high wavenumber rather than a profile settling.

The decay is therefore asymptotically $1/t$ in amplitude and not self similar in shape. What organises the interior of the window is not a fixed profile and we do not identify it. The same mechanism accounts for $c = 0.52$ exhausting resolution at $t = 42$ while still decaying.

7 Linear stability in renormalised time

Set $s = -\log(T - t)$ and $W(y, s) = (T - t)\omega$. Then $\omega = e^s W$ and $ds/dt = e^s$, so $\omega_t = e^{2s}(W + W_s)$, while $\mathcal{R}$ is quadratic and gives $\mathcal{R}(\omega) = e^{2s}\mathcal{R}(W)$. The factors cancel and

$$W_s = \mathcal{R}(W) - W. \quad (9)$$

The profile is exactly a fixed point of (9), and the eigenvalues of $J = D[\mathcal{R}(\cdot) - \cdot](f)$ are growth rates in renormalised time.

Proposition 5 (Forced eigenvalues). $Jf = f$ and $Jf' = 0$.

Proof. Since $\mathcal{R}$ is quadratic, its derivative satisfies $D\mathcal{R}(f)f = 2\mathcal{R}(f) = 2f$, so $Jf = D\mathcal{R}(f)f - f = f$. The second identity is translation invariance: $\mathcal{R}$ commutes with shifts, so differentiating $\mathcal{R}(f(\cdot + \epsilon)) = f(\cdot + \epsilon)$ at $\epsilon = 0$ gives $D\mathcal{R}(f)f' = f'$ and hence $Jf' = 0$. $\square$

The eigenvalue 1 is not an instability. It reflects the freedom to shift the blowup time: replacing T by $T + \epsilon$ gives $W = f\tau/(\tau + \epsilon)$ with $\tau = T - t$, which to first order is $f - \epsilon e^s f$ and

grows like e^s . The eigenvalue 0 is translation. Numerically, $\|Jf - f\|_\infty / \|f\|_\infty = 1.3 \times 10^{-12}$ and $\|Jf'\|_\infty / \|f'\|_\infty = 4.0 \times 10^{-11}$, so Proposition 5 also serves as a sharp check on the assembled Jacobian.

Every remaining eigenvalue at $a = 0.8$ lies in the open left half plane, at $N = 512, 1024$ and 2048 alike. The profile is therefore linearly stable up to the two symmetry directions, which is the precise sense in which the dynamics selects it, and is consistent with a simulation from $\sin x$ reaching it without tuning. The same test applied to profiles taken from simulations at $a = 0.65, 0.70, 0.75, 0.80$ returns stability in every case.

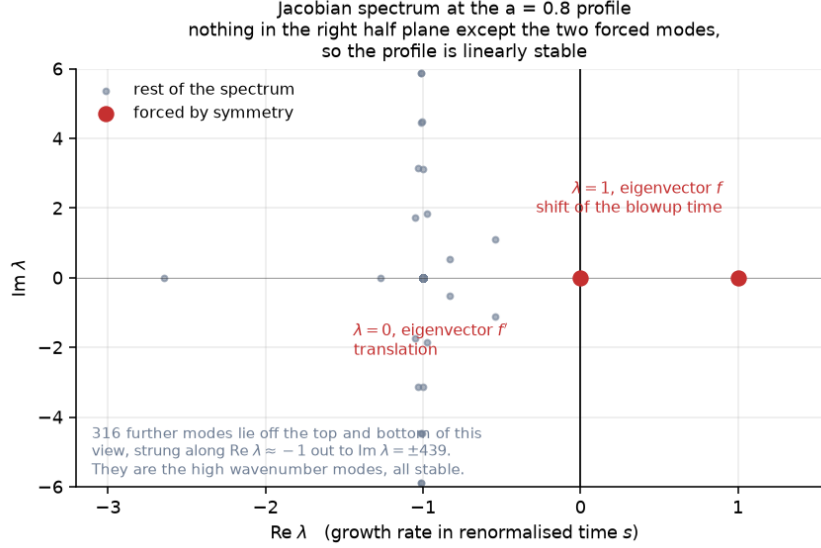


Figure 4: Spectrum of the linearisation (9) about the $a = 0.8$ profile, so that the abscissa is a growth rate in renormalised time s . The two marked eigenvalues are forced by symmetry, at 1 with eigenvector f (freedom to shift the blowup time) and at 0 with eigenvector f' (translation), both verified against Proposition 5 to 10^{-11} or better. Nothing else lies in the closed right half plane. The modes omitted from the view are the high wavenumber ones, strung along $\text{Re } \lambda \approx -1$ and stable.

We note two limitations. The leading stable eigenvalue does not converge in N , taking values $-0.54, -0.146, -0.41$ at $N = 512, 1024, 2048$, so the rate of approach to the profile is not determined here, only its sign. At $a = 0.85$ and $N = 1024$ the Newton solve lands on a different branch, with $\|f\|_\infty = 6.249$ and an unstable eigenvalue at $+2.077$, whereas at $N = 2048$ the same value of a yields $\|f\|_\infty = 5.668$ matching its simulation to 8×10^{-6} . Continuation in a jumps branches outright. The branch structure is real and we have not mapped it.

8 Stagnation points and the local exponent

Rearranging (7) as $aUf' = f(\mathcal{H}f - 1)$ gives

$$\frac{f'}{f} = \frac{\mathcal{H}f - 1}{aU}, \quad (10)$$

so f can only be singular where U vanishes.

Proposition 6 (Local exponent). *Let $U(y_*) = 0$ with $U'(y_*) = c \neq 0$. Then $f(y_*) = 0$, and $f \sim C(y - y_*)^\mu$ with*

$$\mu = \frac{c-1}{ac}. \quad (11)$$

Proof. Evaluating $aUf' = f(\mathcal{H}f - 1)$ at y_* gives $0 = f(y_*)(c-1)$, so $f(y_*) = 0$ unless $c = 1$. Expanding $U(y) = c(y - y_*) + O((y - y_*)^2)$ and using $U' = \mathcal{H}f$, so that $\mathcal{H}f(y_*) = c$, equation (10) becomes $f'/f = \mu/(y - y_*) + O(1)$ with μ as stated, and integrating gives the claim. $\square$

The same computation applies to a narrowing profile. Since the narrowing regime is where most of this family lives, and since it is the case that can be tested against an exact solution, we state it in full rather than as an aside.

Proposition 7 (Local exponent, general self similar form). *Let $\omega = (T - t)^{c_\omega} \bar{\Omega}(x/(T - t)^{c_l})$, so that the profile equation is*

$$(c_l X + a \bar{U}) \bar{\Omega}_X = (c_\omega + \bar{U}_X) \bar{\Omega}, \quad (12)$$

and let X_ be a zero of the total advection velocity $c_l X + a \bar{U}$, which replaces $\bar{U}$ as the transported quantity. Write $h = \mathcal{H}\bar{\Omega}(X_*) = \bar{U}_X(X_*)$ and suppose $c_l + ah \neq 0$. Then $\bar{\Omega}(X_*) = 0$ unless $h = -c_\omega$, and $\bar{\Omega} \sim C(X - X_*)^\nu$ with*

$$\nu = \frac{c_\omega + h}{c_l + ah}. \quad (13)$$

If in addition the zero is simple, that is $\nu = 1$, then

$$h = \frac{c_l - c_\omega}{1 - a}, \quad (14)$$

which for $c_\omega = -1$ reads $h = (1 + c_l)/(1 - a)$.

Proof. Evaluating (12) at X_* annihilates the left hand side, leaving $0 = (c_\omega + h) \bar{\Omega}(X_*)$, so $\bar{\Omega}(X_*) = 0$ unless $h = -c_\omega$. Differentiating the advection velocity gives $(c_l X + a \bar{U})_X = c_l + a \bar{U}_X$, which equals $c_l + ah$ at X_* , so $c_l X + a \bar{U} \sim (c_l + ah)(X - X_*)$ there. Dividing (12) by $\bar{\Omega}$ and by the advection velocity gives $\bar{\Omega}_X/\bar{\Omega} = \nu/(X - X_*) + O(1)$ with ν as stated, and integrating gives the claim. For (14), setting $\nu = 1$ in (13) gives $c_\omega + h = c_l + ah$, that is $h(1 - a) = c_l - c_\omega$. $\square$

Proposition 6 is the case $c_l = 0$, $c_\omega = -1$, which is what a frozen profile requires: no spatial rescaling, and Proposition 8 below is the same case of (14).

Equation (14) is not new. [19] derives $\mathcal{H}\Omega(0) = (1 + c_l)/(1 - a)$ by the same argument, differentiating the profile equation at the stagnation point under the same nondegeneracy $\Omega'(0) \neq 0$, and records the slope $c_l + a \mathcal{H}\Omega(0)$ used in the proof above; we obtained it independently and state it here only because Proposition 6 is the case it does not cover. The content that remains with us is the exponent at a stagnation point where the zero is *not* simple, which is the situation at the second stagnation point on the circle and the one that fixes the spectral decay in Section 10. Section 9 checks the c_l dependence where $c_l \neq 0$ and the answer is known exactly.

Since $|y|^\mu$ has Fourier coefficients decaying like $k^{-(\mu+1)}$, the spectral decay exponent of the profile is

$$\beta = 1 + \frac{c-1}{a c}. \quad (15)$$

One local number therefore predicts the entire spectral decay rate.

Proposition 8 (Exact value at a simple zero). *If in addition f has a simple zero at y_* , that is $f'(y_*) \neq 0$, then*

$$c = \frac{1}{1-a}. \quad (16)$$

Proof. Write $f \sim A(y - y_*)$ with $A = f'(y_*) \neq 0$, $U \sim c(y - y_*)$ and $\mathcal{H}f \sim c$. Matching the coefficient of $(y - y_*)$ in $aUf' = f(\mathcal{H}f - 1)$ gives $a c A = A(c - 1)$, and dividing by A yields $ac = c - 1$, that is $c = 1/(1 - a)$. $\square$

Equivalently $\mu = 1$ at such a point. Numerically, at $N = 4096$, the measured deviation of μ from 1 is -1.7×10^{-15} , -4.6×10^{-14} , -2.0×10^{-10} and 3.5×10^{-8} at $a = 0.70, 0.72, 0.74, 0.76$. The growth of this deviation at larger a tracks the profile's own accuracy, which degrades as the second exponent falls toward one.

Two stagnation points, exactly π apart. Numerically U has exactly two zeros, shown with the profile in Figure 5, and their separation is π to within 9×10^{-16} for every a tested. This is forced. If $U(y_1) = 0$ and $U(y_1 + \pi) = 0$, then expanding both conditions in Fourier modes and adding annihilates the odd modes while subtracting annihilates the even ones, so the even mode part and the odd mode part of U must each vanish at y_1 separately. That is two conditions where a generic zero satisfies one. Direct measurement at $a = 0.8$ gives U_{even} and U_{odd} of order 10^{-15} at both points, against $\|U\|_\infty = 3.48$. We emphasise that this is not the statement that f has only odd modes, which is false: $|\hat{f}_2| = 3315$ against $|\hat{f}_1| = 6520$.

Regularity of the profile. At $a = 0.8$ the second stagnation point has $\mu \approx 2$, and the general solution of (10) near a regular singular point permits different constants on the two sides. We find f'' bounded and exactly odd about y_2 to 10^{-9} , equal to $+2.27045$ on one side and -2.27045 on the other, a jump of 4.54089 with $C_+ = -C_- = 1.13522$. The profile is therefore $C^{1,1}$ but not C^2 . It is not a logarithm: fitting $f'' = 2D \log |y - y_2| + \text{const}$ returns a slope of 0.043 at $R^2 = 0.46$, consistent with zero. Correspondingly f' is continuous at both stagnation points, to 4×10^{-12} and 1.6×10^{-11} .

9 An exact test at $a = 1/2$

Every measurement reported above was taken on a frozen profile, where $c_l = 0$ and (13) cannot be told apart from (11). This section checks the c_l dependence directly, because (2) is solvable in closed form at $a = 1/2$. Nothing in the section is claimed as new: the exact solution is from [18], the exponent $c_l = 1/3$ at $a = 1/2$ is stated in [17] and again in [18], and (14) is from [19]. It is included because the three fit together into a consistency check which is exact rather than numerical, and because the blowup time it produces for our datum does not appear in any of them.

Silantyev, Lushnikov, Siegel and Ambrose [18] construct exact pole dynamics solutions of (2) on the circle for $a = 0$ and $a = 1/2$. In the variable $X = \tan(x/2)$, which maps $(-\pi, \pi)$ onto $\mathbb{R}$, their $a = 1/2$ solution is a conjugate pair consisting of one double and one simple pole, whose amplitudes are tied by $\omega_{-1} = 2iv\omega_{-2}/(1-v^2)$; that constraint is what cancels the logarithmic singularities which advection otherwise generates out of poles. The initial datum used throughout this paper lies in their class, since partial fractions give

$$\sin x = \frac{2X}{1+X^2} = \frac{1}{X-i} + \frac{1}{X+i}, \quad (17)$$

a single conjugate pair at $v = 1$. That is the limit in which the complex singularity sits at infinite height, as it must for an entire datum, and it is the one point at which their parametrisation degenerates: the constraint reads $1 = (2i/0) \cdot 0$ there. The degeneracy is removable, and clearing it gives the following, which is a specialisation of [18] rather than a new solution.

Proposition 9 (Blowup time at $a = 1/2$). *For $a = 1/2$ and $\omega_0 = \sin x$, the solution of (2) blows up at*

$$T = \pi \quad (18)$$

exactly, with the complex singularity reaching the real axis at $x = \pm\pi$.

Proof. The constraint forces $\omega_{-2,i}(0) = (v^2(0) - 1)/(2v(0))$, and substituting this into the coefficient of the pole equation of [18] cancels the vanishing denominator against the vanishing amplitude, leaving $K = 1/[2(v^2(0) + 1)^2]$, which is finite and equal to $1/8$ at $v(0) = 1$. The pole position therefore satisfies the regular equation

$$\frac{dv}{dt} = \frac{(v^2 + 1)^3}{32v^2}, \quad v(0) = 1. \quad (19)$$

Set $G(v) = v(v^2 - 1)/(v^2 + 1)^2 + \arctan v$, so that $G'(v) = 8v^2/(v^2 + 1)^3$ and (19) integrates to $G(v(t)) = G(1) + t/4$. The right hand side of (19) is positive, so v increases and blowup occurs as $v \rightarrow \infty$, which is the pole reaching the real axis at $x = \pm\pi$. Since $G(1) = \pi/4$ and $G(\infty) = \pi/2$, $T = 4[G(\infty) - G(1)] = \pi$. $\square$

Taking $v \rightarrow \infty$ in the closed form near $x = \pi$, with $z = x - \pi$ and $\zeta = vz/2$, and using $dv/dt \sim v^4/32$ so that $v^3 = 32/(3(T-t))$, yields a self similar profile in the sense of Proposition 7 with

$$c_l = \frac{1}{3}, \quad c_\omega = -1, \quad \bar{\Omega}(\zeta) = -\frac{16}{3} \frac{\zeta}{(1 + \zeta^2)^2}, \quad (20)$$

whence $\mathcal{H}\bar{\Omega} = -\frac{8}{3}(\zeta^2 - 1)/(\zeta^2 + 1)^2$ and $\bar{U} = \frac{8}{3}\zeta/(\zeta^2 + 1)$. The value $c_l = 1/3$ is not deduced here for the first time: it is the exponent $\alpha = 1/3$ recorded at $a = 1/2$ in [17], where it also arises as the exact value of their approximation $\alpha_0 = 2/(\gamma(\gamma + 1))$ at $\gamma = 2$, and it is restated in [18] for both of their blowup types. The computation above merely confirms that our datum sits on the same branch. The total advection velocity is

$$c_l\zeta + a\bar{U} = \frac{\zeta}{3} \frac{\zeta^2 + 5}{\zeta^2 + 1}, \quad (21)$$

whose only zero is $\zeta = 0$, the blowup point itself, and $\bar{\Omega}$ has a simple zero there. Proposition 7 then predicts $h = (1 + c_l)/(1 - a) = (4/3)/(1/2) = 8/3$, against $1/(1 - a) = 2$ for a frozen profile. The two differ by a third, so the test discriminates.

Measuring $h = (T - t) \mathcal{H}\omega(\pi, t)$ directly on the exact solution, with $T - t = 4[G(\infty) - G(v)]$ and the same spectral Hilbert transform used everywhere else in this paper, gives Table 6. The residual of the profile equation (12) for (20) is 1.3×10^{-15} , and $v^3(T - t) \rightarrow 32/3$ confirms $c_l = 1/3$.

v	10	30	100	300	1000
h measured	2.671992	2.667259	2.666720	2.666673	2.6666671
$ h - 8/3 $	5.3×10^{-3}	5.9×10^{-4}	5.3×10^{-5}	5.9×10^{-6}	4.5×10^{-7}
$ h - 2 $	6.7×10^{-1}	6.7×10^{-1}	6.7×10^{-1}	6.7×10^{-1}	6.7×10^{-1}

Table 6: The general form of Proposition 7 against the exact $a = 1/2$ solution, at resolutions $N = 2^{14}$ to 2^{20} . The measurement converges on $(1 + c_l)/(1 - a) = 8/3$ and stands off the frozen value $1/(1 - a) = 2$ by two thirds at every resolution.

The same exercise calibrates Table 2. That table measures $c_l = 0.325$ at $a = 0.5$ against the exact $1/3$, a shortfall of 2.5 percent, while at $a = 0$, where $c_l = 1$ exactly, it returns 1.048 to 1.058. The error is therefore a few percent and is not of one sign, which is what the fuller comparison against the published branch in the introduction already showed, and what makes the withdrawn estimate of a_c untenable.

Finally, $\bar{\Omega}$ in (20) is analytic on $\mathbb{R}$, its nearest singularities being at $\zeta = \pm i$. The $C^{1,1}$ regularity and the algebraic spectral decay (15) found in Section 8 are thus features of the frozen regime specifically, not of blowup in this family generally. That is consistent with Proposition 7: with $c_l > 0$ the total advection velocity has a nonzero slope $c_l + ah$ at the stagnation point even when $\bar{U}$ alone would vanish to higher order, so the regular singular point is milder.

10 Determination of the global constant

Proposition 8 pins the stagnation point at which the profile is smooth, and which therefore contributes no singularity. The exponent β is set by the other one, whose c , written c_2 below, is genuinely global: no local balance determines it.

Reading c_2 off a single grid is unreliable. Across $N = 512$ to 4096 it wanders between -1.6486 and -1.6681 without a monotone trend, and a three parameter fit of the form $c_2(N) = c_2^\infty + AN^{-p}$ returns $p = 3.3$ with amplitude 1.1×10^7 , which is a fit to noise. The reason is the $C^{1,1}$ regularity established above: every quantity read from the profile converges algebraically.

The remedy is that c_1 is known exactly. Both constants are read from the same profile and carry the same discretisation error, entering linearly, so c_2 is a linear function of c_1 across resolutions and evaluating that line at the exact c_1 cancels the error without requiring its size or its rate. At $a = 0.8$ the regression over seven resolutions is

$$c_2 = -0.980361 c_1 + 3.240505, \quad R^2 = 0.99999744, \quad (22)$$

and the raw spread of 1.95×10^{-2} collapses to 2.43×10^{-5} , a factor of 801. The corrected value

from $N = 512$ then lies within 1.5×10^{-5} of that from $N = 4096$. We obtain

$$c_2 = -1.66130027 \pm 1.0 \times 10^{-5}, \quad \mu_2 = 2.00242268, \quad \beta = 3.00242268. \quad (23)$$

a	c_2	scatter	μ_2	β
0.70	-0.10252834	4.1×10^{-16}	15.36200	16.36200
0.71	-0.20450843	6.0×10^{-16}	8.29546	9.29546
0.72	-0.31519119	1.4×10^{-14}	5.7953854	6.7953854
0.74	-0.56651088	4.2×10^{-10}	3.7367448	4.7367448
0.76	-0.86560395	1.8×10^{-8}	2.8358720	3.8358720
0.78	-1.22477889	1.3×10^{-6}	2.3288127	3.3288127
0.80	-1.66130509	1.2×10^{-5}	2.0024205	3.0024205
0.82	-2.20024724	5.3×10^{-5}	1.7737736	2.7737736

Table 7: The global constant and the resulting exponents. The scatter is the spread of the corrected values across resolutions and measures self consistency, not residual systematic error. The rows at $a = 0.70$ and $a = 0.71$ were computed over $N = 1024$ to 3072 rather than the full ladder, and are quoted to the digits that run resolves.

Remark 10 ($\beta \neq 3$). The value $\mu_2 = 2$, and hence $\beta = 3$, would require $c_2 = 1/(1 - 2a) = -1.66666667$ at $a = 0.8$. The corrected value misses this by 5.37×10^{-3} , some five hundred times the scatter of the corrected points. Since the correction removes eight hundred times the raw error, the conclusion survives allowing a residual systematic an order of magnitude above the observed scatter. Table 7 shows β varying continuously with a ; the value 3 is merely where that curve crosses.

Remark 11 (Against the published exponent). [17] measure this same exponent for the frozen profile, calling it p_b , by a two domain least squares fit to the spectrum, and report $p_b = 9.32592$ at $a = 0.71$. Equation (15) with c_2 from Table 7 gives $\beta = 9.29546$ there, lower by 0.03046, or 0.33 percent. The two determinations share nothing: theirs is a fit to the tail of the spectrum over a window, ours is a single value of $\mathcal{H}f$ read off the profile at a point. Given the window sensitivity documented in the next remark we read the difference as a correction to the fitted value rather than a disagreement, and we claim no more than the three digits it rests on.

Their exponent diverges as $a \rightarrow a_c^+$, and β here does the same, which gives an independent handle on a_c . Extrapolating $1/\beta$ linearly to zero from the pairs (0.70, 0.71) and (0.71, 0.72) gives 0.6868 and 0.6828, with the pair nearer the transition returning the larger value, pointing at their $a_c = 0.6890665337007457 \dots$ and not at anything larger. This is an extrapolation through convex data, so we quote it for its direction only, but that direction is what withdraws the estimate of a_c discussed in the introduction.

Remark 12 (On measuring β from the spectrum directly). We caution against it. Fitting $|\hat{f}_k| \sim k^{-\beta}$ over a band is sensitive to the band at the 2×10^{-2} level: shifting the window from $128 \leq k \leq 512$ to $0.094 k_{\text{cut}} \leq k \leq 0.375 k_{\text{cut}}$, the same window to three digits, moves the answer from 3.00315 to 2.98626. Across ten reasonable windows a binned estimator spans 2.9829 to 3.0072 and an unbinned least squares fit spans 3.0132 to 3.0510, and the two barely

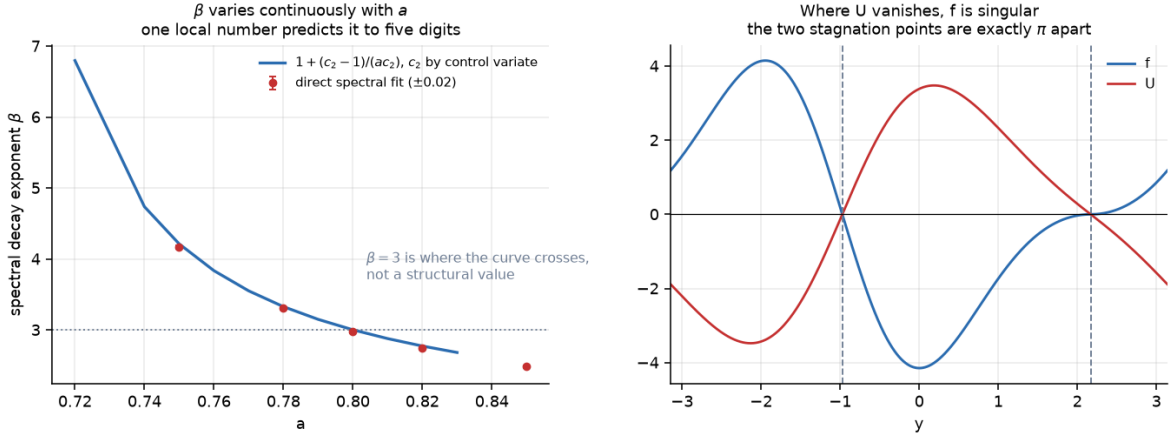


Figure 5: Left: the spectral decay exponent β against a , from (15) with c_2 fixed by the control variate. The curve is smooth and $\beta = 3$ is simply where it crosses. Direct fits of $|\hat{f}_k| \sim k^{-\beta}$ to the spectrum are shown for contrast with error bars of ± 0.02 , which is their window sensitivity; the relation is four orders of magnitude more accurate than measuring the exponent directly. Right: the profile f and its velocity U at $a = 0.8$. The dashed lines mark the two zeros of U , which are exactly π apart, and f vanishes at both as Proposition 6 requires.

overlap. The two ends of the spectrum are biased in opposite directions: the low end by the analytic factor multiplying $|y - y_*|^\mu$, which contributes $k^{-(\mu+2)}$ and faster, and the high end by the dealiasing mask. Refinement distinguishes them, since the former is pinned to absolute k and the latter to a fixed fraction of k_{cut} . Equation (15) with a well determined c_2 is four orders of magnitude more accurate than any of these fits.

11 A first integral and a global constraint

Equation (10) integrates in closed form. On any interval free of zeros of U ,

$$f = C |U|^{1/a} \exp\left(-\frac{1}{a} \int \frac{dy}{U}\right). \quad (24)$$

Near a simple zero this yields $|y - y_*|^{1/a} |y - y_*|^{-1/(ac)} = |y - y_*|^\mu$, recovering Proposition 6 independently.

Proposition 13 (Global constraint). *If f and U are 2π periodic and $\log |f|$ has no jump at either zero of U , then*

$$\text{PV} \oint \frac{dy}{U} = 0. \quad (25)$$

Proof. Integrating (10) once around the circle, $0 = \frac{1}{a} [0 - \text{PV} \oint dy/U]$, since both $\log |f|$ and $\log |U|$ return to their initial values. $\square$

The hypothesis holds here: at y_1 the profile is smooth with $\mu = 1$, and at y_2 the one sided constants satisfy $C_+ = -C_-$, so $|C_+| = |C_-|$ and the logarithm does not jump. We verify

(25) by subtracting both simple poles with $\cot$, which carries the same residue and whose own principal value on the circle vanishes. The resulting ratios are 2.2×10^{-12} , 6.7×10^{-13} , 9.8×10^{-13} and 1.4×10^{-12} at $a = 0.75, 0.78, 0.80, 0.82$, stable across three independent grid shifts.

Remark 14. A naive evaluation of (25) on the grid returns ratios between 10^{-2} and 1 and appears to refute it. This is catastrophic cancellation: the stagnation points land on grid points to within 10^{-11} of a cell, so the subtraction differences two quantities of size 10^8 . Evaluating on a half shifted grid, which is exact for a band limited field, resolves it.

No closed form for c_2 . With c_2 determined across eleven values of a , candidate shapes can be excluded rather than merely left unconfirmed. Every Möbius form is rejected: $\mu_2 = (p + qa)/(1 + ra)$ leaves an rms residual of 1.9×10^{-4} , $c_2 = (p + qa)/(1 + ra)$ leaves 3.0×10^{-4} , and the same shape in the variable $1/(1 - a)$ leaves 6.0×10^{-4} , each one to two orders above the data's precision. Fits of quadratic over quadratic achieve residuals of order 10^{-7} but use five parameters against eleven points and dip below the data scatter at the rough end of the range, so they absorb smooth error rather than signal and we do not regard them as evidence.

We suggest this is the correct answer rather than a gap. Proposition 8 succeeds because it is a *local* balance at a point where f is smooth. No such balance is available at the second stagnation point, so its value is determined by the global solution. Writing ψ for the positive frequency part of U , so that $U = \psi + \bar{\psi}$ and $f = i(\psi' - \bar{\psi}')$, equation (7) reads

$$a(\psi + \bar{\psi})(\psi'' - \bar{\psi}'') = (\psi' - \bar{\psi}')(\psi' + \bar{\psi}' - 1), \quad (26)$$

and the mixed products $\psi\bar{\psi}''$ and $\bar{\psi}\psi''$ straddle both halves of the spectrum. That failure to separate is exactly what makes $a = 0$ integrable and $a \neq 0$ not.

12 The critical limit $a \rightarrow 1$

The case $a = 1$ is degenerate in a way no other member of the family is.

Proposition 15 (Line of equilibria). *At $a = 1$, every $A \sin x$ with $A \in \mathbb{R}$ is a steady state of (2), as is every translate.*

Proof. With $\omega = A \sin x$ we have $\mathcal{H}\omega = -A \cos x$, $u = -A \sin x$ and $\omega_x = A \cos x$, so $\omega \mathcal{H}\omega = -A^2 \sin x \cos x = u \omega_x$ and $\mathcal{R}_1(\omega) = \omega \mathcal{H}\omega - u \omega_x = 0$. $\square$

So the critical case carries a two parameter manifold of equilibria, amplitude and translation, which exists at $a = 1$ and nowhere else. Writing $\varepsilon = 1 - a$, the family is destroyed at order ε and the blowup time diverges. We measure T without fitting: for a frozen profile $\|\omega\|_\infty = \|f\|_\infty / (T - t)$ exactly, so $T = t + \|f\|_\infty / \|\omega\|_\infty$ at any late time, and running to $\|\omega\|_\infty = 10^6$ makes the correction term of order 10^{-6} .

The exponent does not settle: it falls from 0.83 to 0.64 and returns to 0.82 at the smallest ε . No functional form fits the last points, $C\varepsilon^{-p}$, $C/\varepsilon + D$, $C \log(1/\varepsilon) + D$ and $A/\varepsilon + B \log(1/\varepsilon) + C$ leaving rms residuals of 0.99, 0.71, 4.22 and 0.38 against T values from 6 to 57. We therefore quote no exponent. What follows instead is an account of why the natural reductions do not settle it either.

a	0.80	0.90	0.93	0.95	0.96	0.97	0.98	0.99
T	6.0719	10.7569	14.0844	17.7129	20.4591	24.5830	32.1786	56.9742
$T\varepsilon$	1.214	1.076	0.986	0.886	0.818	0.737	0.644	0.570
local p	-	0.828	0.755	0.681	0.646	0.638	0.664	0.824

Table 8: Blowup time as $a \rightarrow 1$, and the local exponent in $T \sim \varepsilon^{-p}$ between consecutive entries. $T\varepsilon$ decreases throughout, so T grows strictly slower than $1/\varepsilon$, but p does not converge.

12.1 The two mode reduction is exact, and gives the wrong answer

Expanding about the manifold, put $\omega = A \sin x + B \sin 2x$. Using $\mathcal{H} \sin kx = -\cos kx$ and $u = -k^{-1} \sin kx$,

$$\begin{aligned}\omega \mathcal{H}\omega &= -\frac{A^2}{2} \sin 2x - AB \sin 3x - \frac{B^2}{2} \sin 4x, \\ u \omega_x &= -\frac{A^2}{2} \sin 2x - \frac{5AB}{4} \sin 3x + \frac{3AB}{4} \sin x - \frac{B^2}{2} \sin 4x,\end{aligned}$$

so that $\mathcal{R}_a = \omega \mathcal{H}\omega - a u \omega_x$ has coefficients $-\frac{3a}{4}AB$ on $\sin x$, $-\frac{\varepsilon}{2}A^2$ on $\sin 2x$, $(\frac{5a}{4} - 1)AB$ on $\sin 3x$ and $-\frac{\varepsilon}{2}B^2$ on $\sin 4x$, each verified against the solver to 10^{-17} .

The $\sin x$ terms cancel in the first line but not the second, so the feedback of mode two into mode one survives at *linear* order. Any argument predicting $p = 2/3$ from a parity cancellation is therefore excluded, not by numerics but by the calculation.

Retaining the two modes gives $A' = -\frac{3a}{4}AB$, $B' = -\frac{\varepsilon}{2}A^2$, which integrates in closed form. Setting $L = \log A$ at $a = 1$ gives $L' = -\frac{3}{4}B$ and $L'' = \frac{3\varepsilon}{8}e^{2L}$, a Liouville equation; multiplying by L' and using $L(0) = L'(0) = 0$ gives $L'^2 = \frac{3\varepsilon}{8}(e^{2L} - 1)$, and since $\int_0^\infty (e^{2L} - 1)^{-1/2} dL = \arccos(e^{-L})|_0^\infty = \pi/2$,

$$T_2 = \frac{\pi}{2} \sqrt{\frac{8}{3\varepsilon}} = 2.56510 \varepsilon^{-1/2}. \quad (27)$$

Against a two mode Galerkin truncation the ratio T/T_2 reads 1.118, 1.054, 1.026, 1.010, 1.005 at $\varepsilon = 0.2, 0.1, 0.05, 0.02, 0.01$. The reduction is exact and gives $p = 1/2$, which is not the behaviour of the full equation.

12.2 The exponent is carried by the cascade

Retaining m modes and measuring the exponent over $0.02 \leq \varepsilon \leq 0.1$ gives the following.

m	2	4	6	8	12	24
p	0.474	0.506	0.518	0.525	0.536	0.554

Table 9: Exponent from an m mode Galerkin truncation. The value climbs monotonically and is still climbing at $m = 24$; at $\varepsilon = 0.1$ the gap to the full answer closes like $m^{-1/2}$.

No finite truncation reproduces the exponent. The case $m = 3$ is degenerate and sits outside the trend, giving $T = 405$ at $\varepsilon = 0.1$ and no blowup at all at 0.05 or 0.02.

12.3 There is no spectral gap, hence no reduction

Both failures have one cause. Since $\mathcal{R}$ is quadratic, $D\mathcal{R}_1(A \sin x) = A M$ with $M = D\mathcal{R}_1(\sin x)$ independent of A , and writing $\omega = A \sin x + v$,

$$v_t = A M v - \frac{\varepsilon}{2} A^2 \sin 2x, \quad A' = -\frac{3}{4} A b_2, \quad (28)$$

with b_2 the $\sin 2x$ component of v . Were M strictly stable, v would saturate at $\frac{\varepsilon}{2} A M^{-1} \sin 2x$ and (28) would reduce to $A' = \kappa \varepsilon A^2$, a Riccati equation giving $p = 1$. That step is unavailable.

Observation 16 (No gap). The spectrum of M is purely imaginary. Computed densely at $N = 128, 256, 512$, every eigenvalue has real part at the 10^{-5} roundoff level against imaginary parts reaching 72, and at $N = 256$ none has positive real part. Its kernel contains the tangent directions to the manifold exactly: $\|M \sin x\|_\infty = 8 \times 10^{-14}$ and $\|M \cos x\|_\infty = 6 \times 10^{-14}$.

Moreover M is not diagonalisable. Integrating $v_t = M v$ from a random perturbation, $\|v\|$ grows linearly, by factors 187, 736, 1392, 2873, 5738 at $t = 10, 50, 100, 200, 400$, doubling as t doubles; with imaginary spectrum this means a Jordan block at zero. Splitting the state identifies its origin. The components along $\sin x$ and $\cos x$ grow linearly, reaching 2663 and -845 at $t = 400$, while the remainder stays bounded and oscillates between 135 and 467. The secular growth is therefore *drift along the manifold of equilibria* rather than instability: a perturbation alters the amplitude and phase at a constant rate, carrying the state to a different steady state rather than away from all of them. The Jordan block is generated by the symmetries themselves, consistent with the stability theorem of Jia, Stewart and Sverak [4], which is stability modulo exactly these symmetries.

We note also that M is strongly non-normal: the remainder above is amplified from 7.8 to 467 before returning, a transient factor near 60 with no eigenvalue to account for it.

The consequence is structural. Modulo the kernel every mode oscillates without decay and the kernel modes drift linearly, so there is no gap anywhere. Hence quasi steady elimination is invalid, because the transient it discards never dies; mode truncation is not a reduction, because the discarded modes never relax, which is why the truncated exponent converges only algebraically; and there is no centre manifold to reduce onto, because the centre subspace is the whole space. The cascade cannot be eliminated because nothing within it is fast relative to anything else, and (27), exact yet giving the wrong exponent, is the sharpest available demonstration that a finite reduction misses the mechanism rather than approximating it.

13 Limitations

The datum $\sin x$ is not generic: it is precisely the ground state of the $a = 1$ model. Table 4 and Section 6 between them answer the worst form of this objection, reaching the same profile from sixteen random data, from a continuous family of offsets and from three designed data, always with the same constants to seven digits.

What remains open is the shape of the basin boundary rather than its existence. Two one parameter families were probed in a space of functions, and the decay window of Section 6 already shows the boundary is not simply whether the datum changes sign. Nothing here maps it in more than one dimension, and the interior of the decay window is not resolved: $c = 0.52$ goes under resolved at $t = 42$ while still decaying.

All statements about $a = 0.9$ and above are unresolved. The residual of the profile equation there does not fall monotonically with the amplitude cap, taking values 0.78, 2.93, 1.77 at caps 10^4 , 10^6 , 10^8 , so those measurements reflect the fitting window rather than the solution. Since $T(0.9) \approx 10.8$ against 6.1 at $a = 0.8$, comparisons should be made at a fixed fraction of T rather than a fixed amplitude.

The scatter quoted in Table 7 measures how consistently the control variate reproduces itself across resolutions. A residual nonlinear dependence of the discretisation error on c_1 would cancel from it and go undetected, so the figure of 1.4×10^{-14} at $a = 0.72$ should not be read as fourteen digit accuracy.

On priority. The bibliographic details of every reference above have been verified. Ten closely related works, [3], [11], [12], [13], [14], [15], [16], [17], [18] and [19], have been read for overlap, and the honest summary is that this paper is a local theory laid on top of a setting which [17] established.

We claim none of the following, all of which are in [17]: the two regime picture on the circle, the critical advection a_c , the frozen blowup $\omega = f(x)/(T - t)$ and its numerical profile, the derivative jump at the antipodal point, the algebraic spectral decay it causes, and the nonlinear stability of the profile as observed from converging simulations. Equation (14) and hence Proposition 7 are in [19]. We derived the relation independently and became aware of [19] only afterwards, which is a statement about how the work went and not a claim on it: the relation is theirs, and we state it here only because Proposition 6 is the case it does not cover. Proposition 8 is its $c_l = 0$ case, and shares its algebra with Theorem 1 of [17]. The exact $a = 1/2$ solution used in Section 9 is from [18] and Proposition 9 is a specialisation of it, claimed only as such.

What we do claim, and did not find in any of them, is Proposition 6 at the stagnation point where the zero of f is not simple, which converts the fitted exponent of [17] into a predicted one; the parity argument fixing the two stagnation points exactly π apart; the control variate determination of c_2 and the resulting exclusion of $\beta = 3$, together with the correction in Remark 11; the spectrum of the linearisation about the frozen profile on the circle; and the first integral and global constraint of Section 11.

This list grew three times while being compiled, most consequentially at the end, and that history is itself a caveat. [17], [18] and [19] belong to a line of work using complex singularity and pole dynamics methods which searches organised around the other references do not surface, and it is the line which works on the circle, the setting this paper relies on. We have read [17] in full and [19] in part, and we expect that literature to bear further on Sections 10 and 9. For the preprints this rests on a reading of the text rather than a line by line verification of every lemma. Both numerical coincidences with [16] have been examined; neither is an overlap, and the first, the exchange of stability near $a \approx 0.75$, is now to be compared against $a_c = 0.68907$ rather than against the withdrawn estimate, and remains unsettled.

The critical limit is left open. Section 12 reports no exponent for $T(\varepsilon)$ because the measured one does not converge over the range we can reach, and reaching $\varepsilon = 10^{-3}$ would need some thirty times the 866,000 steps that $\varepsilon = 10^{-2}$ already took. Observation 16 explains why the shortfall cannot be made up by analysis in place of computation, but it is an explanation of a difficulty rather than a resolution of it.

Finally, nothing here is a proof. A computer assisted proof in this setting, in the style of Chen and Hou [9], requires interval arithmetic producing certified enclosures rather than floating point with convergence studies. The objects this paper produces, a profile equation

with a linearly stable solution and an exact relation fixing one of its two local constants, are the objects such a proof would need to enclose.

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